

Session 1

Preliminaries

Javascript is a high-level, interpreted programming language that is widely used for web development. It was created by Brendan Eich in 1995 and has since become one of the most popular programming languages in the world. JavaScript is primarily used for client-side scripting, allowing developers to create interactive and dynamic web pages. It can also be used on the server-side with environments like Node.js, Deno, Bun and Cloudflare Workers (don't mix with web workers).

In this session, we will introduce the basic JavaScript constructs, conditional statements and control flow statements. The lab manual includes good examples and exercises to practice these concepts.

Suggestion

Before trying the solutions provided for the exercises, try to solve them on your own. This will help you understand the concepts better and improve your problem-solving skills.

Also, Lab manual includes great examples before exercises, make sure to go through them as well.

Objectives

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The objectives of this session are to:

- Understand the basic JavaScript constructs
- Understand conditional statements and control flow statements

Arithmetic Operators

Problem Statement

■ Write in lab record

Q1: Write a JavaScript code to perform the two digit arithmetic operations: Addition, Subtraction, Multiplication and Division.

Approach

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To perform arithmetic operations on two digits, we can follow these steps:

- Define two variables to hold the digits we want to operate on.
- Use the arithmetic operators (`+` , `-` , `*` , `/`) to perform the desired operations.
- Print the results to the console.

Code Implementation

■ Write in lab record

arithmetic-operation.js

```
1 // 1. Define the input numbers
2 // We use 'let' or 'const' to store our numeric values
3 const num1 = 15;
4 const num2 = 10;
5
6 console.log(`Operating on numbers: ${num1} and ${num2}\n`);
7
8 // 2. Addition (+)
9 // The sum of two numbers
10 const addition = num1 + num2;
11 console.log("Addition: " + num1 + " + " + num2 + " = " + addition);
12
13 // 3. Subtraction (-)
14 // Subtracting the second number from the first
15 const subtraction = num1 - num2;
16 console.log("Subtraction: " + num1 + " - " + num2 + " = " + subtraction);
17
18 // 4. Multiplication (*)
19 // The product of two numbers
20 const multiplication = num1 * num2;
21 console.log("Multiplication: " + num1 + " * " + num2 + " = " + multiplication);
22
23 // 5. Division (/)
24 // Dividing the first number by the second
25 // Note: We check if num2 is zero when taking input to avoid mathematical error
26 const division = num1 / num2;
27 console.log("Division: " + num1 + " / " + num2 + " = " + division);
```

Explanation

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- **Variables:** We use `const` (constant) because these values aren't changing during the script's execution. If you want to change them later, you would use `let`.
- **Arithmetic Operators:** JavaScript uses standard mathematical symbols: `+`, `-`, `*`, and `/`.
- **Console.log:** This is the standard method to print output to the debugging console.
- **Division Safety:** In computer science, "Division by Zero" is a common logical error. Including a conditional `if` check demonstrates good programming practice for your lab submission.

- **String Concatenation:** Using the `+` symbol between a string (text in quotes) and a variable allows us to combine them into a single readable sentence for the output.

Pattern Printing

Problem Statement

■ Write in lab record

Q2: Write a JavaScript code to print 1 for 1 time 2 for 2 times, 3 for 3 times and so on up to 10.

Approach

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The goal is to create a visual pattern where each number `n` repeats itself `n` times on a single line. To achieve this, we need Nested Loops:

- **The Outer Loop** will act as a counter that moves from 1 to 10. This tells us which number we are currently processing.
- **The Inner Loop** will run based on the value of the outer loop. If the outer loop is at 5, the inner loop will execute 5 times to print the digit "5".
- We will collect the digits in a string for each row so they stay on the same line, then print that row before moving to the next number.

Code Implementation

■ Write in lab record

pattern-printing.js

```
1 // Outer loop: Determines the number to be printed (1 through 10)
2 for (let i = 1; i ≤ 10; i++) {
3   let row = ""; // Reset the row string for the current number
4
5   // Inner loop: Controls the frequency of the print
6   // It runs 'i' times for the number 'i'
7   for (let j = 1; j ≤ i; j++) {
8     row += i + " "; // Append the number and a space to the row
9   }
10
11  // Print the completed row to the console
12  console.log(row.trim());
13 }
```

Explanation

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- **Nested Loop Logic:** Think of this like a clock. The outer loop is the “hour” hand (moving slowly), and the inner loop is the “minute” hand (completing a full cycle for every single step the hour hand takes).
- **Dynamic Range:** The inner loop condition `j ≤ i` is what makes the pattern triangular. Because the limit of the inner loop depends on the current value of the outer loop, the rows get progressively longer.
- **String Accumulation:** In JavaScript, `console.log()` automatically adds a new line. To get numbers like `2 2` on the same line, we must first build a string (`rowOutput`) and print it once the inner loop finishes its work for that row.
- **trim() Method:** We use `.trim()` at the end to remove the very last space of each row, ensuring the output is clean for your lab record.

Portfolio Exercise

Problem Statement

■ Write in lab record

Q3: Design a web page that describes you. Your page must include the following details (extra details and creativity are welcome):

- Your personal details, i.e. name, gender etc.
- 2-3 lines describing yourself as About Me. Make the most important word bold to emphasize them.
- Details of classes in tabular form you are taking right now (as per your time table).
- Details of your favourite movies, books, or TV shows, in reverse order (using CSS), list at least 3 from each. You can use images, hyperlinks, colourful text to make this section attractive.
- Include a section showing your interest and also not of your interest jobs. In this section also you can use images, showing you're happy in doing your interest jobs and the other to represent you when you're sad.
- In this section show something interesting about one or more people of your neighbours.

Preview

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Make it you

Change the config in the `script` tag to your own name and personal information before you submit.

Code

■ Write in lab record

portfolio.html

```
1 <!doctype html>
2 <html lang="en">
3   <head>
4     <meta charset="UTF-8" />
5     <meta name="viewport" content="width=device-width, initial-scale=1.0
6     <title>Personal Profile Matrix</title>
7     <link rel="preconnect" href="https://fonts.googleapis.com" />
8     <link rel="preconnect" href="https://fonts.gstatic.com" crossorigin
9     <link
10       href="https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans
11       rel="stylesheet"
12   />
13
14   <style>
15     /* All core colors are bound to CSS variables controlled by the
16     :root {
17       --bg-main: #0b0f19;
18       --bg-card: #131a26;
19       --bg-card-hover: #1c2636;
20       --border: #223147;
21       --text-primary: #f3f4f6;
22       --text-secondary: #9ca3af;
23
24       --accent-primary: #ff3e00;
25       --accent-secondary: #00b894;
26       --accent-muted: #3b82f6;
27       --accent-danger: #ef4444;
28
29       --font-sans:
30         "Plus Jakarta Sans", -apple-system, BlinkMacSystemFont,
31         sans-serif;
32       --font-mono: "JetBrains Mono", monospace;
33     }
34
35     * {
36       margin: 0;
37       padding: 0;
38       box-sizing: border-box;
39     }
40     body {
41       font-family: var(--font-sans);
42       background-color: var(--bg-main);
43       color: var(--text-primary);
```

```
44         min-height: 100vh;
45         padding: 40px 20px;
46         line-height: 1.6;
47         background-image:
48             radial-gradient(
49                 circle at 10% 20%,
50                 rgba(255, 62, 0, 0.03) 0%,
51                 transparent 40%
52             ),
53             radial-gradient(
54                 circle at 90% 80%,
55                 rgba(59, 130, 246, 0.03) 0%,
56                 transparent 40%
57             );
58     }
59
60     .container {
61         max-width: 1100px;
62         margin: 0 auto;
63     }
64     header {
65         text-align: center;
66         margin-bottom: 50px;
67     }
68     h1 {
69         font-size: 3.5rem;
70         font-weight: 800;
71         letter-spacing: -0.03em;
72         margin-bottom: 12px;
73         color: #ffffff;
74     }
75     .subtitle {
76         font-family: var(--font-mono);
77         font-size: 1rem;
78         color: var(--accent-primary);
79         font-weight: 500;
80     }
81
82     .bento-grid {
83         display: grid;
84         grid-template-columns: repeat(3, 1fr);
85         gap: 24px;
86         margin-bottom: 40px;
87     }
```

```
88     .card {
89         background-color: var(--bg-card);
90         border: 1px solid var(--border);
91         border-radius: 16px;
92         padding: 30px;
93         transition: all 0.3s cubic-bezier(0.4, 0, 0.2, 1);
94     }
95     .card:hover {
96         border-color: rgba(255, 255, 255, 0.1);
97         transform: translateY(-2px);
98     }
99     h2 {
100
101         font-size: 1.35rem;
102
103         font-weight: 700;
104
105         margin-bottom: 20px;
106
107         display: flex;
108
109         align-items: center;
110
111         gap: 10px;
112     }
113
114     .col-span-2 {
115
116         grid-column: span 2;
117     }
118
119     .col-span-3 {
120
121         grid-column: span 3;
122     }
123
124     .detail-grid {
```

```
11
 6      display: grid;
11
 7      grid-template-columns: repeat(2, 1fr);
11
 8      gap: 16px;
11
 9  }
12
10  .detail-label {
12
11      color: var(--text-secondary);
12
12      font-size: 0.85rem;
12
13      text-transform: uppercase;
12
14      font-family: var(--font-mono);
12
15      display: block;
12
16      margin-bottom: 2px;
12
17  }
12
18
12
19  .table-container {
13
20      width: 100%;
13
21      overflow-x: auto;
13
22      margin-top: 10px;
13
23  }
13
24  table {
13
25      width: 100%;
13
26      border-collapse: collapse;
13
27      font-size: 0.9rem;
```

```
13
8      text-align: left;
13
9    }
14
0    th {
14
1      background-color: rgba(255, 255, 255, 0.03);
14
2      font-family: var(--font-mono);
14
3      color: var(--accent-muted);
14
4      font-weight: 600;
14
5      text-transform: uppercase;
14
6      padding: 12px 16px;
14
7      border-bottom: 2px solid var(--border);
14
8    }
14
9    td {
15
0      padding: 14px 16px;
15
1      border-bottom: 1px solid var(--border);
15
2    }
15
3    .course-code {
15
4      font-family: var(--font-mono);
15
5      color: var(--accent-primary);
15
6      font-weight: 600;
15
7    }
15
8
15
9    .favorites-container {
```

```
16
0      display: grid;
16
1      grid-template-columns: repeat(3, 1fr);
16
2      gap: 16px;
16
3      width: 100%;
16
4  }
16
5  .fav-column h3 {
16
6      font-size: 0.9rem;
16
7      font-family: var(--font-mono);
16
8      margin-bottom: 16px;
16
9      text-transform: uppercase;
17
10 }
17
1
17
2  /* CSS Flex rule handles structural list inversion naturally */
17
3  .reverse-list {
17
4      display: flex;
17
5      flex-direction: column-reverse;
17
6      gap: 12px;
17
7  }
17
8  .favorite-item {
17
9      background-color: rgba(255, 255, 255, 0.02);
18
10     border: 1px solid var(--border);
18
11     padding: 16px;
```

```
18
2      border-radius: 10px;
18
3    }
18
4    .favorite-item a {
18
5      color: var(--text-primary);
18
6      text-decoration: none;
18
7      font-weight: 600;
18
8      font-size: 1rem;
18
9      display: block;
19
10     margin-bottom: 4px;
19
11   }
19
12   .favorite-item a:hover {
19
13     color: var(--accent-muted);
19
14   }
19
15   .favorite-item p {
19
16     color: var(--text-secondary);
19
17     font-size: 0.85rem;
19
18   }
19
20
21   .career-split {
20
22     display: grid;
20
23     grid-template-columns: 1fr 1fr;
20
24     gap: 20px;
```

```
20
 4      }
20
 5      .career-col {
20
 6          background-color: rgba(255, 255, 255, 0.01);
20
 7          border: 1px dashed var(--border);
20
 8          border-radius: 12px;
20
 9          padding: 20px;
21
10          position: relative;
21
11      }
21
12      .career-title-love {
21
13          color: var(--accent-secondary);
21
14          margin-bottom: 16px;
21
15          font-size: 1.1rem;
21
16      }
21
17      .career-title-avoid {
21
18          color: var(--accent-danger);
21
19          margin-bottom: 16px;
22
20          font-size: 1.1rem;
22
21      }
22
22      .status-avatar {
22
23          position: absolute;
22
24          top: 15px;
22
25          right: 20px;
```

```
22
6      font-family: var(--font-mono);
22
7      font-size: 1.4rem;
22
8      font-weight: bold;
22
9    }
23
0    .love .status-avatar {
23
1      color: var(--accent-secondary);
23
2    }
23
3    .avoid .status-avatar {
23
4      color: var(--accent-danger);
23
5    }
23
6
23
7    .job-node {
23
8      display: flex;
23
9      align-items: flex-start;
24
0      gap: 12px;
24
1      margin-bottom: 14px;
24
2      background: rgba(255, 255, 255, 0.02);
24
3      padding: 12px;
24
4      border-radius: 8px;
24
5    }
24
6    .job-node .dot {
24
7      margin-top: 6px;
```

```
24
8      width: 6px;
24
9      height: 6px;
25
0      border-radius: 50%;
25
1    }
25
2    .career-col.love .dot {
25
3      background-color: var(--accent-secondary);
25
4    }
25
5    .career-col.avoid .dot {
25
6      background-color: var(--accent-danger);
25
7    }
25
8    .job-meta strong {
25
9      font-size: 0.95rem;
26
0      display: block;
26
1    }
26
2    .job-meta p {
26
3      font-size: 0.85rem;
26
4      color: var(--text-secondary);
26
5    }
26
6
26
7    .fun-fact {
26
8      margin-top: 14px;
26
9      background-color: rgba(255, 255, 255, 0.02);
```

```
27
0      border-left: 3px solid var(--accent-muted);
27
1      padding: 12px 16px;
27
2      border-radius: 0 8px 8px 0;
27
3      font-size: 0.9rem;
27
4  }
27
5  footer {
27
6      margin-top: 60px;
27
7      border-top: 1px solid var(--border);
27
8      padding-top: 30px;
27
9      text-align: center;
28
0      font-family: var(--font-mono);
28
1      font-size: 0.8rem;
28
2      color: var(--text-secondary);
28
3  }
28
4
28
5  @media (max-width: 900px) {
28
6      .bento-grid {
28
7          grid-template-columns: 1fr;
28
8      }
28
9      .col-span-2,
29
0      .col-span-3 {
29
1          grid-column: span 1;
```

```
29
2      }
29
3      .favorites-container {
29
4          grid-template-columns: 1fr;
29
5      }
29
6      .career-split {
29
7          grid-template-columns: 1fr;
29
8      }
29
9      }
30
0      </style>
30
1      </head>
30
2      <body>
30
3          <div class="container">
30
4              <!-- Structural targets that JavaScript hooks into and populates
30
5              <header id="header-root"></header>
30
6
30
7          <div class="bento-grid">
30
8              <div class="card col-span-3" id="about-root"></div>
30
9              <div class="card col-span-3">
31
0                  <h2><span>📄</span> Personal Profile Specifications</h2>
31
1                  <div class="detail-grid" id="profile-root"></div>
31
2              </div>
31
3              <div class="card col-span-3" id="timetable-root"></div>
```

```
31
4      <div class="card col-span-3" id="favorites-root"></div>
31
5      <div class="card col-span-2" id="careers-root"></div>
31
6      <div class="card" id="neighbor-root"></div>
31
7    </div>
31
8
31
9    <footer id="footer-root"></footer>
32
0  </div>
32
1
32
2    <!-- CENTRAL CONTROL MATRIX: Change everything below in the CONFIG o
32
3    <script>
32
4      const CONFIG = {
32
5        // Theme Customization Engine
32
6        theme: {
32
7          backgroundMain: "#0b0f19",
32
8          backgroundCard: "#131a26",
32
9          borderLine: "#223147",
33
0          textMain: "#f3f4f6",
33
1          textMuted: "#9ca3af",
33
2
33
3          accentSvelte: "#ff3e00", // Primary accent (Svelte Orang
33
4          accentGreen: "#00b894", // Success accent
33
5          accentBlue: "#3b82f6", // Informational accent
```

```
33
6      accentRed: "#ef4444", // Warning/Sad accent
33
7      },
33
8
33
9      // Personal Information
34
0      personal: {
34
1          name: "Shivam Meena",
34
2          tagline: "~/fullstack-engineer/mca-student-ignou",
34
3          aboutHTML: `I'm a seasoned **full-stack software engineer
34
4          metaDetails: [
34
5              { label: "Identity Name", value: "Shivam Meena" },
34
6              { label: "Gender Configuration", value: "Male" },
34
7              {
34
8                  label: "Communications Target",
34
9                  value: "meenashivam823@gmail.com",
35
0              },
35
1              {
35
2                  label: "Secure Phone Matrix",
35
3                  value: "+91 7303043479",
35
4              },
35
5              {
35
6                  label: "Academic Location",
35
7                  value: "IGNOU Matrix Center",
```

```
35
 8      },
35
 9      {
36
10          label: "Academic Track Pipeline",
36
11          value: "Master of Computer Applications (MCA) –
36
12      },
36
13  ],
36
14  },
36
15
36
16  // Live Timetable Data
36
17  timetable: {
36
18      workerURL:
36
19          "https://mca-class-notifier.ethercorps.workers.dev/"
37
20      slots: [
37
21          "Session slot 1 (09:00 - 11:00)",
37
22          "Session slot 2 (11:30 - 13:30)",
37
23          "Session slot 3 (14:30 - 16:30)",
37
24      ],
37
25      days: [
37
26          {
37
27              day: "Monday",
37
28              classes: [
37
29                  '<span class="course-code">MCS-211</span>: C
```

```
38
0      '<span class="course-code">MCS-212</span>: D
38
1      "🔬 Practical Lab Session I",
38
2      ],
38
3      },
38
4      {
38
5      day: "Tuesday",
38
6      classes: [
38
7      '<span class="course-code">MCS-213</span>: S
38
8      '<span class="course-code">MCS-211</span>: C
38
9      "📖 Self-Directed Research",
39
0      ],
39
1      },
39
2      {
39
3      day: "Wednesday",
39
4      classes: [
39
5      '<span class="course-code">MCS-212</span>: D
39
6      '<span class="course-code">MCS-214</span>: D
39
7      "🔬 Practical Lab Session II",
39
8      ],
39
9      },
40
0      {
40
1      day: "Thursday",
```

```
40
2          classes: [
40
3              '<span class="course-code">MCS-211</span>: C
40
4              '<span class="course-code">MCS-213</span>: S
40
5              "📖 Open Seminar Loop",
40
6          ],
40
7      },
40
8      {
40
9          day: "Friday",
41
10         classes: [
41
11             '<span class="course-code">MCS-214</span>: D
41
12             '<span class="course-code">MCS-212</span>: D
41
13             "🐭 Systems Evaluation Lab",
41
14         ],
41
15     },
41
16 ],
41
17 },
41
18 // Media Favorites Indices
42
19 favorites: {
42
20     movies: [
42
21         {
42
22             title: "1. RRR",
```

```
42
4         desc: "An absolute visual masterpiece of scale a
42
5         link: "https://www.imdb.com/title/tt8178634/",
42
6     },
42
7     {
42
8         title: "2. Mission: Impossible",
42
9         desc: "High-stakes algorithmic planning and exec
43
0         link: "https://www.imdb.com/title/tt0120755/",
43
1     },
43
2     {
43
3         title: "3. Suzume",
43
4         desc: "Spatial thresholds and closure dynamics."
43
5         link: "https://www.imdb.com/title/tt13320662/",
43
6     },
43
7 ],
43
books: [
43
8     {
44
9         title: "1. C For Dummies",
44
0         desc: "A foundational structural text for master
44
1         link: "https://www.amazon.com/C-Dummies-Alan-Sim
44
2
3     },
44
4     {
44
5         title: "2. The Rust Book",
```

```
44
6         desc: "The definitive structural shifts covering
44
7         link: "https://www.rust-lang.org/learn",
44
8     },
44
9     {
45
10         title: "3. Berserk",
45
11         desc: "Kentaro Miura's legendary peak dark fanta
45
12         link: "https://www.goodreads.com/book/show/20527
45
13     },
45
14 ],
45
15 shows: [
45
16     {
45
17         title: "1. Psycho-Pass",
45
18         desc: "An intricate psychological deep dive into
45
19         link: "https://www.imdb.com/title/tt1467304/",
46
20     },
46
21     {
46
22         title: "2. Person of Interest",
46
23         desc: "A prophetic cyberpunk analysis regarding
46
24         link: "https://www.imdb.com/title/tt2049116/",
46
25     },
46
26     {
46
27         title: "3. Fullmetal Alchemist: Brotherhood",
```

```
46
8         desc: "The absolute gold standard highlighting s
46
9         link: "https://www.imdb.com/title/tt0421357/",
47
10        },
47
11    ],
47
12    },
47
13
47
14    // Career Alignment Mapping
47
15    careers: {
47
16        happyEmoji: "^___^",
47
17        sadEmoji: "T___T",
47
18        interested: [
47
19            {
48
20                title: "Systems Programmer",
48
21                desc: "Compiling highly performant engine toolin
48
22            },
48
23            {
48
24                title: "Reactive UI Architect",
48
25                desc: "Crafting lightning-fast reactivity via Sv
48
26            },
48
27            {
48
28                title: "Fullstack Systems Developer",
48
29                desc: "Assembling robust, optimized end-to-end c
```

```
49
0          },
49
1          ],
49
2          disinterested: [
49
3              {
49
4                  title: "Outbound Sales Agent",
49
5                  desc: "Cold pitching generic software layers ove
49
6              },
49
7              {
49
8                  title: "Cold Outreach Specialist",
49
9                  desc: "Spamming template sequences with near zer
50
0              },
50
1              {
50
2                  title: "Metric-Chasing Sales Roles",
50
3                  desc: "Prioritizing raw volume metrics over high
50
4              },
50
5          ],
50
6      },
50
7
50
8      // Neighbors Node
50
9      neighbor: {
51
0          name: "Dr. [Name]",
51
1          role: "Geologist PhD",
```

```
51
2      context: "Fascinating cross-disciplinary contact point."
51
3      fact: "We are currently discussing techniques to feed ma
51
4      },
51
5  };
51
6
51
7  // --- PIPELINE COMPILATION RENDER ENGINE ---
51
8  function runEngine() {
51
9      // Apply Theme Palette Colors
52
10     const root = document.documentElement.style;
52
11     root.setProperty("--bg-main", CONFIG.theme.backgroundMain);
52
12     root.setProperty("--bg-card", CONFIG.theme.backgroundCard);
52
13     root.setProperty("--border", CONFIG.theme.borderLine);
52
14     root.setProperty("--text-primary", CONFIG.theme.textMain);
52
15     root.setProperty("--text-secondary", CONFIG.theme.textMuted);
52
16     root.setProperty("--accent-primary", CONFIG.theme.accentSvel
52
17     root.setProperty(
52
18         "--accent-secondary",
52
19         CONFIG.theme.accentGreen,
53
20     );
53
21     root.setProperty("--accent-muted", CONFIG.theme.accentBlue);
53
22     root.setProperty("--accent-danger", CONFIG.theme.accentRed);
53
23
```

```

53
4      // Render Header
53
5      document.getElementById("header-root").innerHTML = `
53
6      <h1>${CONFIG.personal.name}</h1>
53
7      <p class="subtitle">${CONFIG.personal.tagline}</p>
53
8      `;
53
9
54
0      // Render About (and parse Markdown stars dynamically)
54
1      const parsedAbout = CONFIG.personal.aboutHTML.replace(
54
2          /\*\*(.*?)\*\*/g,
54
3          "<strong>$1</strong>",
54
4      );
54
5      document.getElementById("about-root").innerHTML = `
54
6      <h2><span>👤 </span> About Me</h2>
54
7      <p style="color: var(--text-secondary); font-size: 1.15rem;
54
8      `;
54
9
55
0      // Render Profile Parameters Matrix
55
1      document.getElementById("profile-root").innerHTML =
55
2          CONFIG.personal.metaDetails
55
3              .map(
55
4                  (d) => `
55
5      <div class="detail-item">

```

```

55         <span class="detail-label">${d.label}</span>
6
55         ${d.value}
7
55     </div>
8
55     `
9
56     ,
0
56     )
1
56     .join("");
2
56
3     // Render Timetable
56
4     document.getElementById("timetable-root").innerHTML = `
56
5     <h2><span>📅</span> Academic Class Timetable (Semester 1)</h2>
56
6     <div class="table-container">
56
7         <table>
56
8             <thead>
56
9                 <tr>
57
0                     <th>Day Sequence</th>
57
1                     ${CONFIG.timetable.slots.map((s) => `<th>${s
57
2                 </tr>
57
3             </thead>
57
4             <tbody>
57
5                 ${CONFIG.timetable.days
57
6                 .map(
57
7                 (d) => `

```

```

57
8          <tr>
57
9          <td><strong>${d.day}</strong></td>
58
0          ${d.classes.map((c) => `<td>${c}</td>`).
58
1          </tr>
58
2          `,
58
3          )
58
4          .join("")}
58
5          </tbody>
58
6          </table>
58
7      </div>
58
8      <p style="margin-top: 15px; font-size: 0.85rem; color: var(-
58
9          * Sync engines actively handled via edge workers. Source
59
0      </p>
59
1      `;
59
2
59
3      // Helper for Favorites Column Assemblies
59
4      const makeFavList = (arr) =>
59
5          arr
59
6          .map(
59
7              (item) => `
59
8          <div class="favorite-item">
59
9          <a href="${item.link}" target="_blank">${item.title}</a>

```

```
60         <p>${item.desc}</p>
61     0
62         <p>${item.desc}</p>
63     1
64     </div>
65 2
66     `
67 3
68     )
69 4
70     .join("");
71 5
72 // Render Favorites
73 6
74 document.getElementById("favorites-root").innerHTML = `
75 7
76 <h2><span>★ </span> Media & Reference Indices (Rendered in R
77 8
78 <div class="favorites-container">
79 9
80     <div class="fav-column">
81 0
82         <h3 style="color: var(--accent-danger)">🎬 Motion Pi
83 1
84         <div class="reverse-list">${makeFavList(CONFIG.favor
85 2
86     </div>
87 3
88     <div class="fav-column">
89 4
90         <h3 style="color: var(--accent-primary)">📖 Compilat
91 5
92         <div class="reverse-list">${makeFavList(CONFIG.favor
93 6
94     </div>
95 7
96     <div class="fav-column">
97 8
98         <h3 style="color: var(--accent-muted)">📺 Broadcast
99 9
100        <div class="reverse-list">${makeFavList(CONFIG.favor
101 0
102    </div>
103 1
```

```

62
2      </div>
62
3      `;
62
4
62
5      // Render Career Alignment Matrices
62
6      document.getElementById("careers-root").innerHTML = `
62
7      <h2><span>🏠 </span> Job Alignment Matrices</h2>
62
8      <div class="career-split">
62
9          <div class="career-col love">
63
10             <div class="status-avatar">${CONFIG.careers.happyEmo
63
11             <h3 class="career-title-love">Target Interests</h3>
63
12             ${CONFIG.careers.interested
63
13                 .map(
63
14                     (j) => `
63
15                     <div class="job-node">
63
16                         <div class="dot"></div>
63
17                         <div class="job-meta"><strong>${j.title}</st
63
18                     </div>
63
19                 `,
64
20             )
64
21             .join("")}
64
22         </div>
64
23         <div class="career-col avoid">

```

```

64
4      <div class="status-avatar">${CONFIG.careers.sadEmoji
64
5      <h3 class="career-title-avoid">Non-Target Paths</h3>
64
6      ${CONFIG.careers.disinterested
64
7          .map(
64
8              (j) => `
64
9              <div class="job-node">
65
0                  <div class="dot"></div>
65
1                  <div class="job-meta"><strong>${j.title}</st
65
2                  </div>
65
3              `,
65
4              )
65
5              .join("")}
65
6          </div>
65
7      </div>
65
8      `;
65
9
66
0      // Render Neighbor Node Analytics
66
1      document.getElementById("neighbor-root").innerHTML = `
66
2      <h2><span>🏠</span> Neighbor Node Analytics</h2>
66
3      <div class="neighbor-card">
66
4          <div class="neighbor-info">
66
5              <p style="font-size: 0.95rem;"><strong style="color:

```

```
66
6      <p style="color: var(--text-secondary); font-size: 0
66
7      <div class="fun-fact">
66
8          <strong>Interdisciplinary Node Connection:</stro
66
9      </div>
67
0      </div>
67
1      </div>
67
2      `;
67
3
67
4      // Render Footer
67
5      document.getElementById("footer-root").innerHTML = `
67
6      <p>© 2026 ${CONFIG.personal.name} | <span style="color: var(
67
7      <p style="opacity: 0.5; font-style: italic;">"Building the f
67
8      `;
67
9      }
68
0
68
1      // Initialize Engine Execution
68
2      runEngine();
68
3      </script>
68
4      </body>
68
5      </html>
```

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